Advanced Java– syllabus
Duration: 80 hours

Topics:

Introduction to Java

- Basics of SQL and DDL,DML statements in SQL, Oracle 11g installation
- What is JDBC
- JDBC Driver types, Database Data types
- Steps to connect to in JDBC
- Transaction management,
- Exception Handling,
- Batch Processing
- Exercise on SQL and JDBC.

HTML (Hyper Text Markup Language)

- How to create your first HTML page, Tags and Attributes
- Paragraph ,Style and Formatting, Text items, Creating tables
- Creating links and lists, Creating backgrounds, Forms in HTML

Exercise on HTML.

XML(Extensible Markup Language)

- XML Fundamentals, XML Document Structure,
- XML Parsers, XML DTDs, XMLSchema

Introduction to Web application

- Building and deploying Web application,
- Introduction to HTTP
- Basics of Servlet, Servlet Life Cycle, Types Of Servlets
- Servlet Request and Response, Servlet Handling, Database Connectivity in Servlet
- Exception handling
- Session handling,
- Page redirection,
- Uploading Files,
- Autorefreshing
- Exercise on Servlets.

Introduction to JSP

- How to create your first JSP.
- Components of JSP – Scriplets,
- Expression and declaration,

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Implicit Objects,
Action elements,
HTML Forms,
MVC architecture
Session handling,
JSP-Auto refresh
Overview, Variables, Operators
Objects ,Control Flow Statements

Functions ,Error Handling, Event handling, Validations

JavaScript Window, JavaScript Documents.

Exercise in JSP and Javascript.

Project on JSP/JavaScript/Servlets/JDBC

Spring Boot Introduction

Spring Boot Version
Spring vs Spring Boot vs Spring MVC
SB Advantage
Spring Boot Architecture
Spring Boot Annotations

How to create a basic application in Java Spring Boot

Best Practices For Structuring Spring Boot Application

Spring Boot – Hello World Example

How to Create a Spring Boot Project in Spring Initializr and Run it in IntelliJ IDEA?
Dynamic Dropdown From Database using Spring Boot
How to Get the Body of Request in Spring Boot?
How to Make Put Request in Spring Boot?
Spring Boot – Code Structure
Spring Boot – RestTemplate
Spring Boot – Change Port
Spring Boot – Scheduling
Spring Boot – Transaction Management Using @Transactional Annotation
Spring Boot – Map Entity to DTO using ModelMapper
Spring Boot – Validation using Hibernate Validator
Spring Boot – Cache Provider
Spring Boot – Logging
Spring Boot – Auto-configuration
Spring Boot – File Handling

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Spring Boot – Groovy Test Introduction

- Spring Boot – Exception Handling
- Spring Boot – Project Deployment Using Tomcat
- Spring Boot – Packaging

Spring Boot – Thymeleaf with Example

- Spring Boot – DevTools
- Spring Boot –Difference Between @Service Annotation and @Repository Annotation
- How to Make a Simple RestController in Spring Boot?

How to Implement Simple Authentication in Spring Boot?

- What is PathVariable in the Spring Boot?
- What is Command Line Runner Interface in Spring Boot?
- Difference Between Spring Boot Starter Web and Spring Boot Starter Tomcat
- How to Dockerize a Spring boot application with Maven
- Creating a Spring Boot App using Dockerfile

Spring Boot with REST API

Spring Boot with Database

Spring Boot CRUD Operation

Spring Boot Microservices

Real time project on Spring Boot

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